ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

Practice Examination (Self-Study)

Course No.: Phy 4143 | Time: 2 Hours

Course Title: Physics II | Full Marks: 100

Scope: Lecture 1–3 only (Basic Concepts, Basic Laws, Y-Δ Transformation)

There are 3 (three) questions. Answer all questions. The symbols have their usual meanings. Marks of each part are written in brackets. (This practice paper describes circuits in words since it was generated for self-study, not handed out in class.)

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1. Theory and Definitions

a) Define voltage and current. State Ohm's Law with its formula. [3]

b) Define an electric circuit. State Kirchhoff's Current Law (KCL) and Kirchhoff's Voltage Law (KVL). [4]

c) Define node, junction, branch, loop and mesh. Clearly explain the difference between a node and a junction. [5]

d) Differentiate between an ideal independent voltage source and an ideal independent current source, including their internal resistance. [4]

e) State the condition under which source transformation is valid, and write the transformation formula in both directions (voltage source to current source, and vice versa). [4]

f) State the rule for combining resistors in series and in parallel. Write the formula for R_eq of n series resistors and of two parallel resistors. [4]

g) State the Voltage Divider Rule and the Current Divider Rule with their formulas, clearly defining every symbol used. [4]

h) Briefly explain, with an example situation, when a Y-Δ (or Δ-Y) transformation becomes necessary in resistive network analysis. [2]

2. Basic Laws — Ohm's Law, KVL, KCL, Source Transformation

a) A 30 V source is connected in series with three resistors R1 = 2 Ω, R2 = 3 Ω and R3 = 5 Ω (a single loop). Find the loop current and the voltage across each resistor. Verify your result using KVL. [8]

b) Two voltage sources, 18 V and 6 V, both oriented to aid each other, drive a single-loop series circuit containing a 3 Ω and a 5 Ω resistor. Find the loop current and the voltage across each resistor. [8]

c) At a single node a, four branches are connected in parallel, each returning to ground: an 8 A independent current source entering node a; a 4 Ω resistor carrying current io (defined flowing from node a to ground); a dependent current source of value io/2 (same reference direction as io); and an 8 Ω resistor with voltage vo defined across it (+ at node a). Use KCL at node a to find io and vo. [10]

d) A practical voltage source consists of a 20 V ideal source in series with a 5 Ω resistor. (i) Find its equivalent current-source representation. (ii) This equivalent current source (with its 5 Ω in parallel) now feeds a 15 Ω load resistor. Use the current divider rule to find the current through the load, and the voltage across it. [9]

3. Resistor Connections, Dividers and Y-Δ Transformation

a) Between terminals A and B: a 4 Ω resistor is in series with the parallel combination of a 6 Ω and a 3 Ω resistor, and this whole combination is in series with a 5 Ω resistor. Find R_eq between A and B. [6]

b) Three resistors R1 = 4 Ω, R2 = 6 Ω, R3 = 10 Ω are connected in series across a 40 V source. Using the Voltage Divider Rule, find the voltage across R2. [6]

c) Three resistors R1 = 6 Ω, R2 = 12 Ω, R3 = 4 Ω are connected in parallel, and a total current of 18 A enters the parallel combination. Using the Current Divider Rule, find the current through R1. [7]

d) A Y-connected (star) resistor network has R1 = 4 Ω, R2 = 6 Ω, R3 = 12 Ω (R1, R2, R3 connect the star point to terminals a, b, c respectively). Convert this network to its equivalent Δ (delta) network, i.e. find R_A, R_B and R_C. [8]

e) Using the Δ network you obtained in part (d), convert it back to a Y network and verify that you recover the original values of R1, R2 and R3. [8]

Answer Key

(Final answers only — try every question first before checking.)

Q2.

a) i = 3 A, v1 = 6 V, v2 = 9 V, v3 = 15 V (sum = 30 V, checks out)

b) i = 3 A, v(3Ω) = 9 V, v(5Ω) = 15 V

c) io = 4 A, vo = 16 V

d) (i) Is = 4 A with 5 Ω in parallel (ii) i(load) = 1 A, v(load) = 15 V

Q3.

a) R_eq = 11 Ω (6Ω parallel 3Ω = 2Ω, then 4+2+5)

b) v2 = 12 V

c) i1 = 6 A (R_parallel = 2 Ω)

d) R_A = 36 Ω, R_B = 24 Ω, R_C = 12 Ω

e) R1 = 4 Ω, R2 = 6 Ω, R3 = 12 Ω — matches the original Y values, checks out